1. Display the count of the number of rows in a table.

```sql
select Noanimals=count(*) from Animal
```

00 %

Results | Messages

	Noanimals
1	26

2. Delete a specified record in a table and prove that it is deleted.

```sql
select * from Animal
where AnimalID = 27
```

100 %

Results | Messages

	AnimalID	Name	Desc	ClassID	HabitatID
1	27	house	NULL	NULL	NULL

```sql
select * from Animal
where AnimalID = 27

delete from Animal
where AnimalID = 27

select * from Animal
where AnimalID = 27
```

100 %

Results | Messages

	AnimalID	Name	Desc	ClassID	HabitatID

3. Two table join that creates a view.

```sql
-- two table join with view
create view vw_getanimalclass as
Select animalname, classname
from Animal, Class
where animal.classid = class.classid
```

100 %

Messages

```
Commands completed successfully.

Completion time: 2025-06-23T18:43:15.2898095-04:00
```

```sql
select * from vw_getanimalclass
```

100 %

Results | Messages

	animalname	classname
1	Black Bear	Mammal
2	Rattlesnake	Reptile
3	Coyote	Mammal
4	Whale	Mammal

4. Three table join qualified with a key value from **one view** and two tables.

```sql
select V.animalname, rangename, className
from animal A, AnimalRange AR, Range R,
    vw_getanimalclass2 V
where A.AnimalID = AR.AnimalID
and AR.RangeID = R.RangeID
and A.ClassID = V.classID
```

100 %

Results | Messages

	animalname	rangename	className
1	Black Bear	Temperate Forest	Mammal
2	Black Bear	Wetlands	Mammal
3	Black Bear	Desert	Mammal
4	Black Bear	Temperate Forest	Mammal
5	Black Bear	Mountain Range	Mammal
6	Black Bear	Wetlands	Mammal
7	Black Bear	Temperate Forest	Mammal

5. Use five SQL functions (such as SUM). This should be five queries with at least one function for each query and not the same functions as you used in previous assignments.

6. Create a SQLstored proccedure to insert values into a row using variables passed in on execution.

```sql
-- SQLstored procedure to insert values into a row using variables passed in on execution.
create proc spu_getanimalname as
SELECT AnimalName
from Animal
```

100 %

Messages

```
Commands completed successfully.

Completion time: 2025-06-23T19:30:42.9808116-04:00
```

```sql
exec spu_getanimalname
```

100 %

Results | Messages

	AnimalName
1	Black Bear
2	Rattlesnake
3	Coyote
4	Whale
5	Walrus

7. Write the SQL to create a table, key, and index. This must not be generated SQL. Example after #12.

```sql
--  create a table, key, and index.
CREATE TABLE student
(studentID int NOT NULL,
FName nvarchar(20) NOT NULL,
LName nvarchar(30) NOT NULL,
MName nvarchar(20) NULL,
CONSTRAINT PK_student PRIMARY KEY CLUSTERED
(studentID ASC)
ON [PRIMARY]) -- key
ON [PRIMARY] -- index
```

100 %

Messages

```
Commands completed successfully.

Completion time: 2025-06-23T19:44:21.0896841-04:00
```

8. Create a query that uses Group By

```sql
select C.classname, nbrincount=count(C.classid)
from animal A, class C
where A.ClassID = C.ClassID
group by C.ClassName
```

100 %

Results | Messages

	classname	nbrincount
1	Amphibian	2
2	Bird	7
3	Fish	2
4	Invertabrates	1
5	Mammal	13
6	Reptile	1

9. Demonstrate the use of distinct, ascending, descending in a query. The result set must have at least 7 rows.

```sql
-- order descending
select animalname
from animal
order by animalname desc
```

100 %

Results | Messages

	animalname
1	Whale
2	Walrus
3	Turkey
4	Timber Wolf
5	Shark
6	Screech Owl
7	Rattlesnake

```sql
-- order ascending
select animalname
from animal
order by animalname asc
```

100 %

Results | Messages

	animalname
1	Barn Owl
2	Barred Owl
3	Black Bear
4	Buffalo
5	Cheetah
6	Chicken
7	Clown Fish

10. Create a query that demonstrates an inner join (includes join in statement)

```sql
-- inner join
select animalname, habitatname
from Animal A, Habbitat H
where A.HabitatID = H.habitatID
```

100 %

Results | Messages

	animalname	habitatname
1	Black Bear	forest
2	Coyote	forest
3	Whale	forest
4	Walrus	forest
5	Rattlesnake	Desert
6	Polar Bear	Arctic

11. Create a query that demonstrates a left outer join (includes join in statement)

```sql
-- left outer join
select animalname, habitatname
from Animal A
	left join Habbitat H
on A.HabitatID = H.habitatID
```

100 %

Results | Messages

	animalname	habitatname
1	Black Bear	forest
2	Rattlesnake	Desert
3	Coyote	forest
4	Whale	forest
5	Walrus	forest

12. Create a query that demonstrates a right outer join (includes join in statement)

```sql
-- right outer join
select animalname, habitatname
from Animal A
	right join Habbitat H
on A.HabitatID = H.habitatID
```

100 %

Results | Messages

	animalname	habitatname
1	Black Bear	forest
2	Coyote	forest
3	Whale	forest
4	Walrus	forest
5	NULL	Savanna
6	Rattlesnake	Desert
7	NULL	Tundra